



Biotechnology and its Applications

1 - INTRODUCTION

Biotechnology: Essentially deals with industrial scale production of biopharmaceuticals and biologicals using GM microbes, fungi, plants and animals.

Applications of biotechnology include:

- Therapeutics
- Processed food
- Diagnostics - Bioremediation
- Genetically modified crops for agriculture
- Waste treatment
- Energy production

Three critical research areas of biotechnology:

- Providing best catalyst in the form of improved microbes or pure enzymes
- Creating optimal conditions through genetic engineering
- Downstream processing technologies for purification

2 - BIOTECHNOLOGICAL APPLICATIONS IN AGRICULTURE

Food production could possibly be increased by three ways:

- Agrochemical based agriculture
- Organic agriculture
- Genetically engineered crop-based agriculture

Green revolution resulted in tripling of food production: Reasons for success of green revolution:

- Improved crop varieties
- Agrochemicals (fertilisers + pesticides)
- Better management practices

Problem Area and Hinderances:

- Enhancement in food production by green revolution was still not enough to feed growing population
- **Agrochemicals** are often too **expensive** for farmers of **developing world**
- **Increase in yield** with existing varieties is **not possible** using **conventional breeding**

Genetically Modified Organisms

Organisms including plants, bacteria, fungi and animals whose **genes have been altered by manipulation** are called **genetically modified organisms (GMO)**.

Applications of genetic modification:

- Made crops more tolerant to abiotic stresses (cold, drought, salt, heat).
- Reduced reliance on chemical pesticides (pest-resistant crops).
- Helped to reduce post harvest losses.
- Increased efficiency of mineral usage by plants (prevents early exhaustion of fertility of soil).
- Enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice.

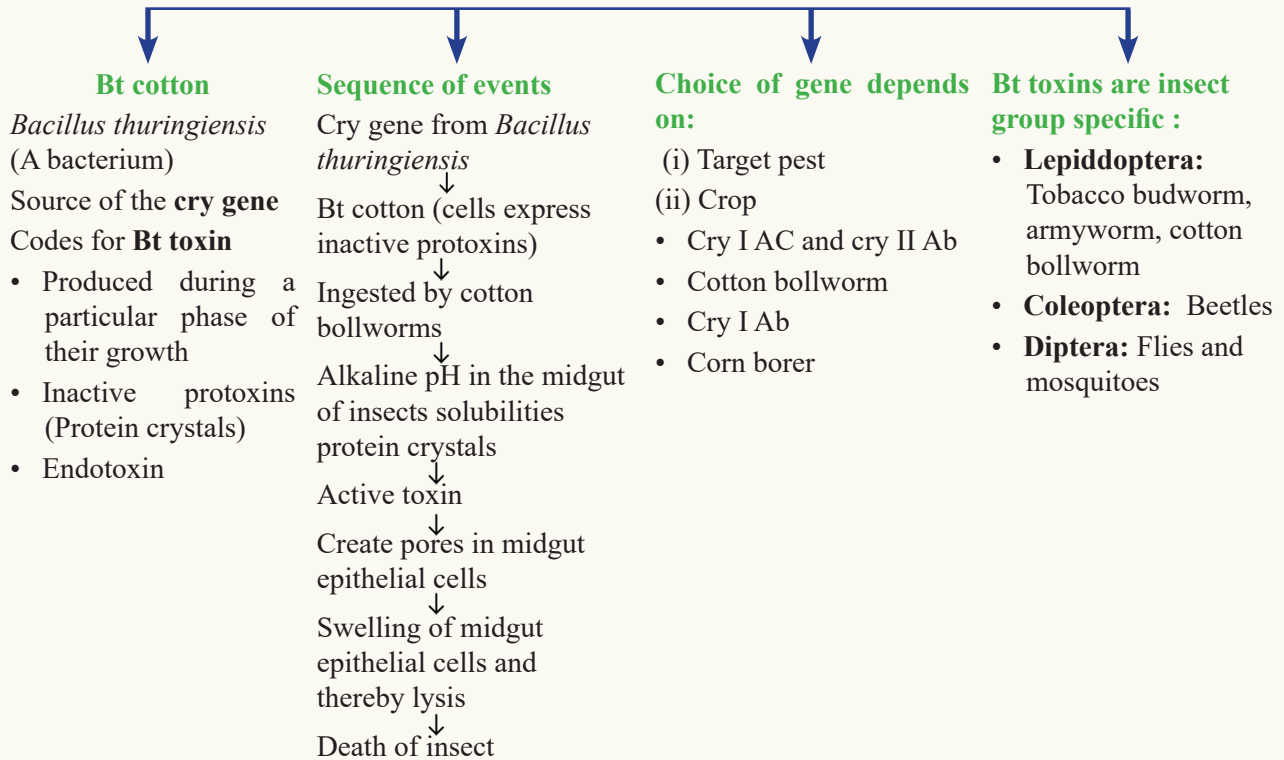
Tailor Made Plants:

Plants have been developed to supply alternative resources to industries in the form of starches, fuels and pharmaceuticals.



3 - INSECT RESISTANT PLANTS

- Provides resistance to insects without the need for insecticides (bio-pesticide)
- Examples of biopesticides are Bt cotton, Bt corn, rice, tomato, potato and soyabean etc.



4 - PEST RESISTANT PLANTS

- Method of cellular defense seen in all eukaryotes against pest infestation.
- Technique responsible : RNA interference (RNAi)
- Based on post transcriptional silencing of mRNA
- Translation of mRNA coded from pest specific genes is silenced/prevented due to formation of complementary dsRNA

Source

I. Viruses with RNA genome

II. Mobile genetic elements Transposons replicating via an RNA intermediate

The case of nematode resistant transgenic tobacco:

- Pest causing **roots knot** disease in tobacco plant : *Meloidogyne incognita* (Nematode/helminth)
- **Nematode specific gene** is introduced in host plant (tobacco), by using Ti plasmid (vector) of *Agrobacterium tumefaciens*, in such a manner that it produces both sense and **antisense RNA** in the host cells.
- Sense RNA and antisense RNA being complementary form **dsRNA** that initiates **RNAi**.
- Parasite could not survive in a **transgenic host** expressing **specific interfering RNA**.
- **Host plant** – generated **dsRNA** triggers protection against nematode infestation.



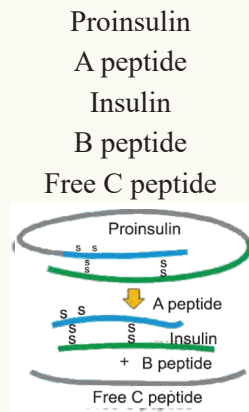
5 - BIOTECHNOLOGICAL APPLICATIONS IN MEDICINE

Advantages of Recombinant Therapeutics:

- 30 recombinant therapeutics have been approved for human use the world over. In India, 12 of these are presently being marketed.
- Mass production of safe and effective drugs.
- Do not induce unwanted immunological responses.

Genetically Engineered Human Insulin

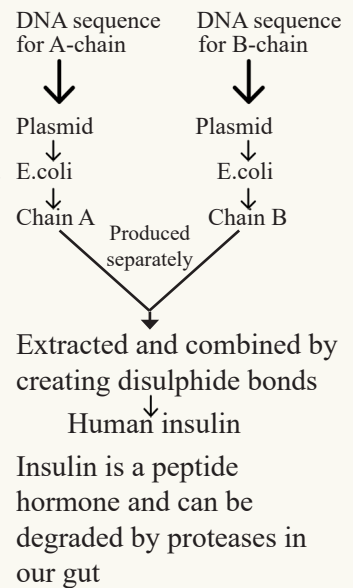
- Problem : Insulin extracted from slaughtered cattle and pigs could cause allergy.
- **Solution:** Production of humulin



Maturation of proinsulin to insulin

- Recombinant insulin manufactured by Eli Lilly, an American company, in 1983
- The main challenge for production of insulin using rDNA techniques was getting insulin assembled into a mature form

Sequence of events: Artificially synthesised



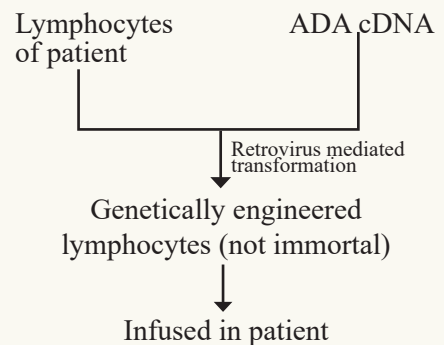
6 - GENE THERAPY

- Insertion of genes into an individual's cells to treat diseases by
 - (i) Replacing a defective mutant allele with a functional one
 - (ii) Gene targeting which involves gene amplification.
- **First clinical gene therapy** was conducted in **1990** in a **4 year old** girl to treat **adenosine deaminase (ADA) deficiency**. ADA enzyme is crucial for immune system to function

Treatment for ADA Deficiency

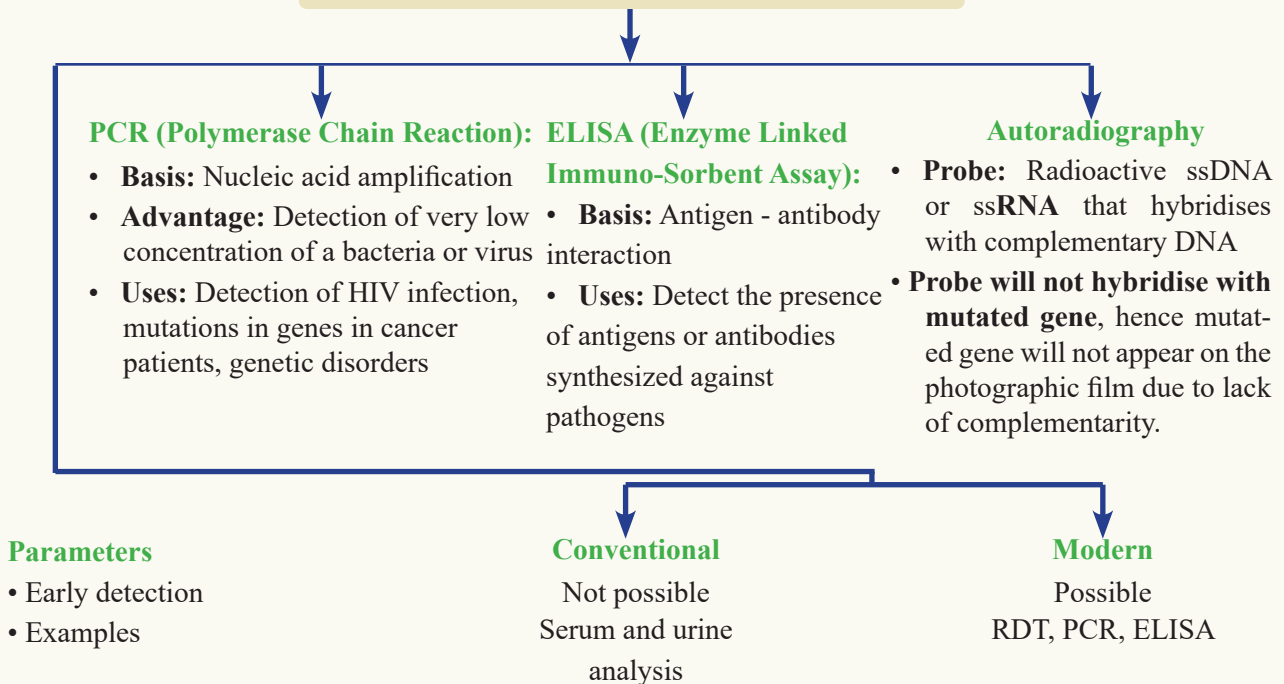
1. Enzyme replacement therapy
 - Functional ADA is given by injection
2. Bone marrow transplantation in children
3. Gene therapy - Could be a permanent cure if bone marrow transplantation is done at early embryonic stages.

Steps in gene therapy

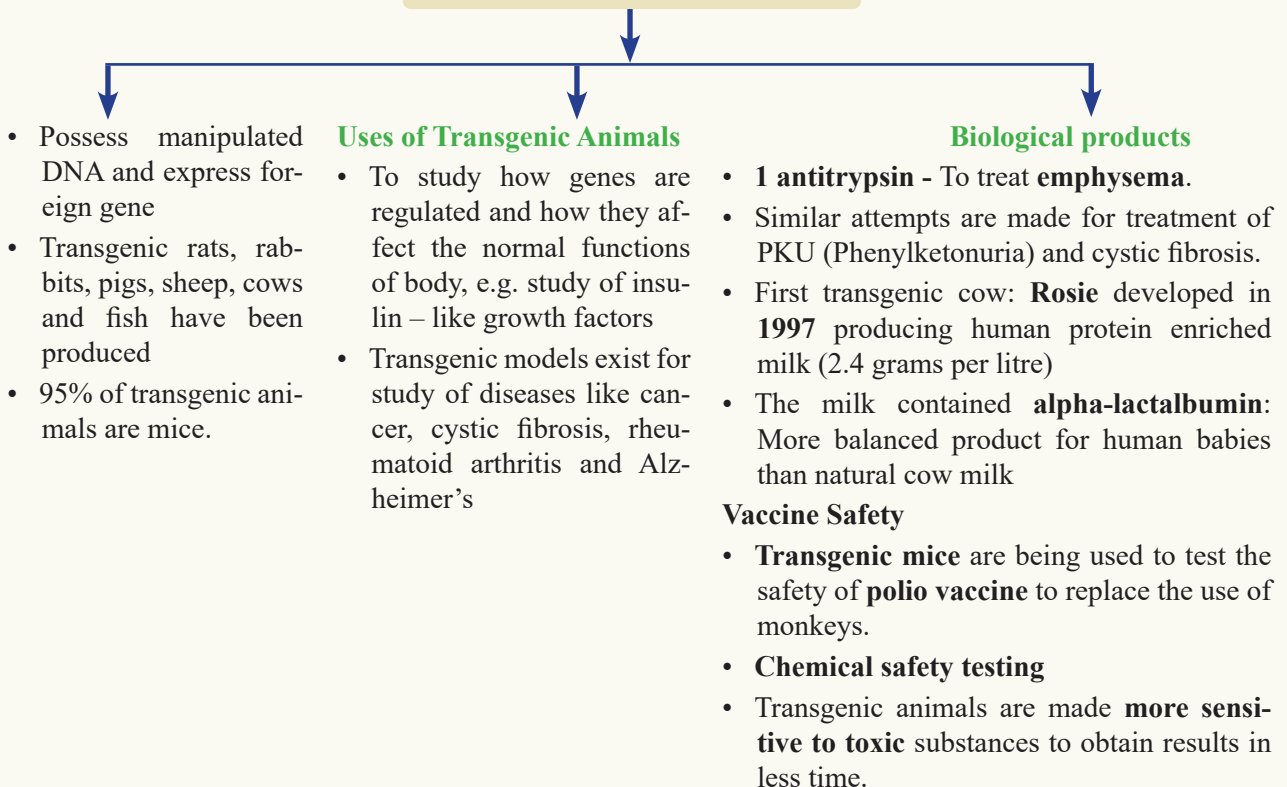




7 - MOLECULAR DIAGNOSIS METHODS



8 - TRANSGENIC ANIMALS





9 - ETHICAL ISSUES

Genetic modification of organisms (GMO) can have unpredictable results when such organisms are introduced into the ecosystem.

- Genetic manipulation of living organisms by humans has to be regulated for moral and biological significance.
- GEAC (Genetic Engineering Approval Committee) : Makes decisions regarding the validity of GM research and the safety of introducing GMO for public services
- Developing countries are rich in biodiversity and traditional knowledge related to bio-resources
- Biopiracy : Refers to the use of bio-resources by multinational companies and other organisations without proper authorization from the countries and people concerned without compensatory payment.

Controversies regarding patents and biopiracy:

(i) Basmati rice :

- 2,00,000 varieties of rice in India. 27 documented varieties of Basmati rice in India
- In 1997, an American company got patent rights on Basmati rice through the US patent and Trade-mark office.

Basmati rice × Semi-dwarf variety of rice



New variety of Basmati rice

(ii) Turmeric

(iii) Neem

The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill.